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Chiral sulfide and achiral sulfonic acid cocatalyzed enantioselective electrophilic tandem selenylation semipinacol rearrangement of allenols

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A highly enantioselective electrophilic selenylation/semipinacol rearrangement of allenols has been developed, which is enabled by the cooperative catalysis of a chiral sulfide and an achiral sulfonic acid. The designed and synthesized chiral sulfide catalyst and selenylating reagent play a crucial role in enhancing both enantioselectivity and reactivity. This approach exhibits excellent regio-, chemo-, and enantioselectivity, providing access to diverse enantioenriched cyclopentanones featuring an arylselenovinyl-substituted quaternary carbon stereocenter. Furthermore, these products can be transformed into synthetically valuable alkyne, vinyl bromide, and aniline derivatives. Mechanistic studies reveal that the combination of a chiral sulfide and an achiral sulfonic acid not only facilitates the formation of catalytically active species, but also governs the enantioselectivity of the reaction. Meanwhile, density functional theory calculations disclose that four hydrogen bond interactions and a π···π interaction are responsible for the observed enantioselectivity.

Rearrangement reactions play a significant role in the total synthesis of natural products and pharmaceuticals due to their efficient skeletal editing capability[1–4]. Consequently, studies on rearrangement reactions, particularly catalytic asymmetric ones, have garnered significant interest among synthetic chemists. In recent decades, there has been a surge in research on catalytic asymmetric electrophilic addition/semipinacol rearrangement of allylic alcohols, leading to the introduction of various electrophilic species into alkenes with high enantioselectivity. A series of works have been achieved by Tu[5–7], Yeung[8,9], Alexakis[10,11], and others[12–16] (Fig. 1a). In contrast, less attention has been given to catalytic asymmetric electrophilic addition/semipinacol rearrangement of allenol substrates, which can provide access to products containing both alkenes and quaternary carbon stereocenters simultaneously. Ma's group has made substantial contributions in this area through their dedicated work on allene chemistry[17–19]. They reported an enantioselective bromination/semipinacol rearrangement of 2,3-allenols in 2014 (Fig. 1b)[20]. To our knowledge, this is the only report on enantioselective electrophilic addition/semipinacol rearrangement of allenols[21,22]. However, only moderate enantioselectivity was achieved (up to 78% ee), indicating that achieving highly enantioselective electrophilic addition/semipinacol rearrangement of allenols remains challenging and desirable.

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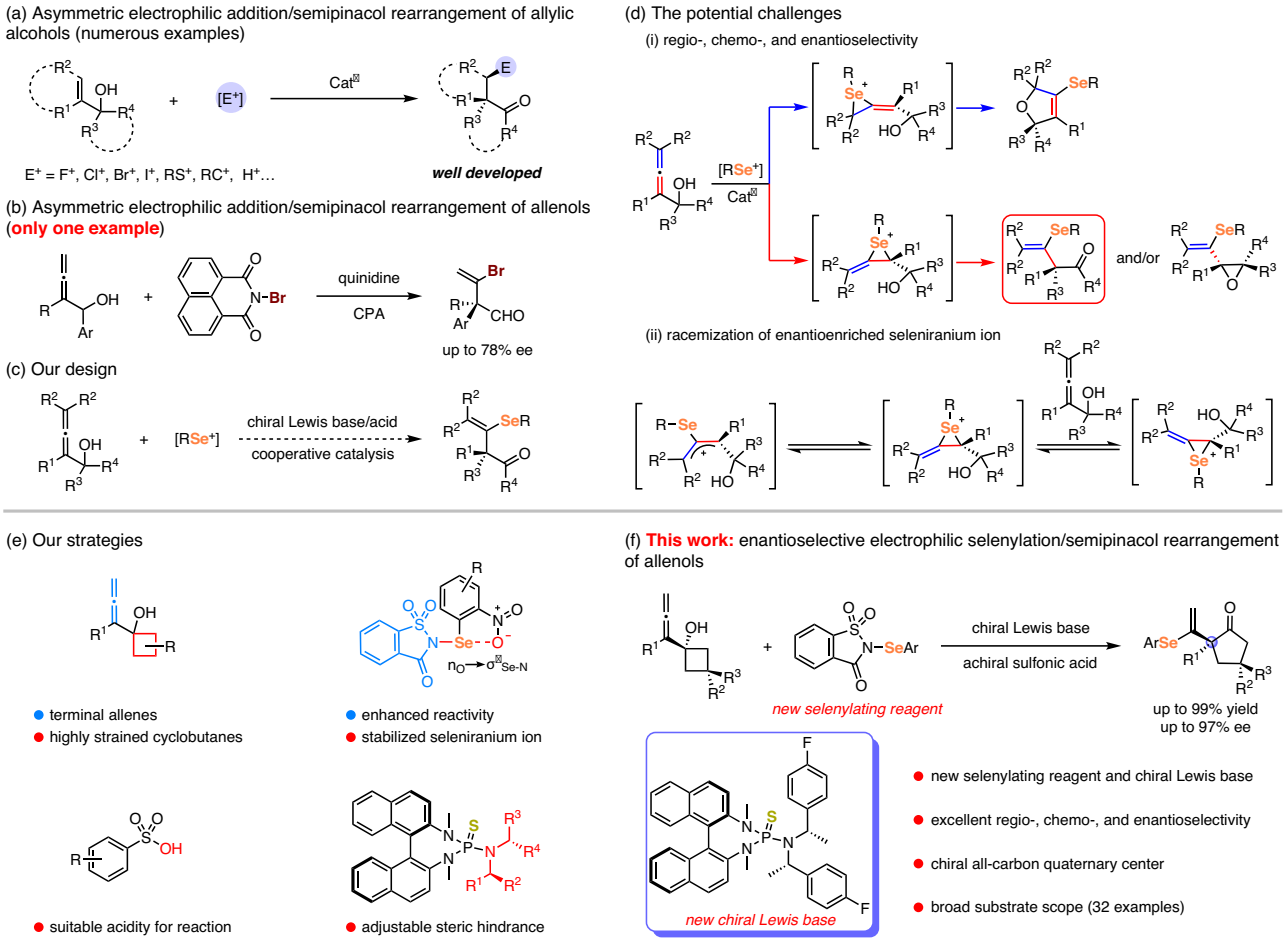


Fig. 1 | Design of catalytic asymmetric electrophilic selenylation/semipinacol rearrangement of allenols. a Asymmetric electrophilic addition/semipinacol rearrangement of allylic alcohols. **b** Asymmetric electrophilic addition/semipinacol rearrangement of allenols. **c** Our design. **d** The potential challenges. **e** Our strategies. **f** This work: enantioselective electrophilic selenylation/semipinacol rearrangement of allenols.

Chiral organoselenium compounds have received increasing attention from researchers due to their unique chemical and pharmacological properties[23,24]. They are not only employed as synthons, catalysts, and ligands in organic synthesis[25–27] but also find applications in biological, medicinal, and material sciences[28,29]. Consequently, several elegant methods for their catalytic asymmetric synthesis have been reported[30–39]. However, there is relatively limited literature on the catalytic asymmetric electrophilic selenofunctionalization of unsaturated substrates due to the facile racemization of enantioenriched seleniranium ion intermediates, which provides a direct approach for synthesizing chiral organoselenium compounds. Recently, notable studies on the catalytic asymmetric electrophilic selenofunctionalization of alkenes have been documented by Denmark[40], Jacobsen[41], Yeung[42,43], Chen[44], and others[45,46]; however, there is no report on the catalytic asymmetric electrophilic selenofunctionalization of allenes.

In this context, in conjunction with our ongoing research in chiral sulfur and selenium chemistry[44,47–51], we envision the potential establishment of electrophilic selenylation/semipinacol rearrangement of allenols with high enantioselectivity (Fig. 1c). This presents significantly greater complexity and difficulty compared to allylic alcohol substrates, primarily due to several factors (Fig. 1d). First, there is a regioselectivity issue when the electrophilic selenium source reacts with the allenol substrate. Second, prioritization of 1,2-shift or nucleophilic attack of hydroxyl oxygen in the subsequent ring-opening step poses a chemoselectivity challenge. Third, achieving high enantioselectivity is difficult due to minor differentiation between the enantiotopic faces of the allene moiety. Furthermore, additional challenges such as facile racemization of enantioenriched seleniranium ions by transferring them from one allene to another and/or delivering them to π-allyl cation intermediates should be considered[52]. To address these challenges, we propose the following strategies (Fig. 1e): (1) terminal allenols with highly strained cyclobutanol moiety could be considered as substrates. For an allenol, its nonterminal C–C double bond is more electron-rich, which would resolve regioselectivity; meanwhile, a highly strained cyclobutanol moiety is conducive to a 1,2-shift, which would resolve the chemoselectivity issue. (2) A saccharin-derived selenylating reagent featuring a 2-nitrophenyl group can be utilized as an electrophilic selenium source. The presence of a nitrophenyl group can stabilize the seleniranium ion intermediate while the saccharin skeleton enhances reactivity inhibiting the racemization process and improving enantioselectivity[40,53–55]. (3) A mild cooperative catalysis of Lewis base and acid is required that could prevent allenol decomposition and enhance enantioselectivity[56–59].

Herein, we report a highly enantioselective electrophilic selenylation/semipinacol rearrangement of allenols enabled by the cooperative catalysis of a chiral sulfide and an achiral sulfonic acid. Additionally, saccharin-derived selenylating reagents and chiral BINAM-derived sulfides are synthesized and utilized (Fig. 1f). A range of enantioenriched cyclopentanones bearing an arylselenovinyl-substituted quaternary carbon stereocenter are obtained, which can be readily transformed into other valuable molecules. Control experiments and density functional theory calculations demonstrate

the essential role of chiral sulfide and achiral sulfonic acid in achieving high reactivity and enantioselectivity.

Results

Reaction condition optimization

Our initial experiment commenced with allenyl cyclobutanol **2a** as the model substrate and *N*-(2-nitrophenylseleneyl)succinimide **3a** as the selenylating reagent (Table 1). After an initial screening, the corresponding selenium-containing cyclopentanone **4** was produced in 40% yield and with 81% enantiomeric excess (ee) using (*R*)-**1a** as the catalyst, *p*-toluenesulfonic acid (pTSA) as the acid, chloroform ($CHCl_3$) as the solvent, and 5 Å molecular sieve (5 Å MS) as the additive at −10 °C under argon (entry 1; for details, please refer to Supplementary Information). The relatively low reactivity of **3a** posed a significant obstacle to improving enantioselectivity further, prompting us to consider changing the core skeleton of selenylating reagent. Unfortunately, no reaction occurred in the presence of *N*-(2-nitrophenylseleneyl)phthalimide **3b** due to its poor solubility in $CHCl_3$ (entry 2). In contrast, *N*-(2-nitrophenylseleneyl)saccharin reagent showed promising results by providing a nearly quantitative yield and slightly improved enantiomeric excess (entry 3). Lowering the temperature to −40 °C increased enantioselectivity to a level of 84% ee with satisfactory yield (entry 4). Subsequently, we carefully investigated an array of chiral BINAM-derived catalysts. Unfortunately, (*R*)-**1b** was decomposed under current reaction conditions leading to only trace amounts of product formation (entry 5). Based on our previous research findings, (*R*)-**1c** featuring a bulky *N*-tert-butyl group was anticipated to enhance enantioselectivity[48,51]. However, in this case, it only achieved 32% yield despite maintaining the same 84% ee (entry 6), highlighting the importance of adjusting the steric hindrance of amine moiety. A family of C_2-and pseudo-C_2-symmetrical secondary amines was prepared and installed on catalysts (entries 7–12)[60,61], after comprehensive comparison, catalyst (*R*, *S*, *S*)-**1i** yielded optimal results with 80% yield and 86% ee (entry 12). Subsequently, the selenylating reagent **3c** was further modified. To our delight, such modification resulted in a significant enhancement to 92% ee with 94% yield (entry 13). Further enhancement to 94% ee was achieved when using 2-naphthalenesulfonic acid (2-NSA) instead of pTSA (entry 14). Diastereoisomer (*S*, *S*, *S*)-**1i** had negligible impact on the ee value (−93% ee vs. 94% ee) without compromising the yield, proving that the absolute configuration of the product **4** is controlled by the BINAM domain of catalyst (entry 15). Thus, reaction conditions presented in entry 14 of Table 1 were deemed optimal for further substrate scope investigation.

Substrate scope

With the optimal reaction conditions in hand, we next explored the reaction scope of this reaction with various allenols and selenylating reagents (Fig. 2). First, aromatic groups of the 2-position of allenols **2** were evaluated, and the majority of them were converted to arylselenovinyl-substituted cyclopentanones with excellent yields and high enantioselectivities (**4a**–**4m**). Electron-donating *p*-tolyl, *p*-phenylphenyl, and 2-naphthyl allenols have a limited influence on the yields and enantioselectivities (**4b**–**4d**). However, the *p*-methoxyphenyl group reduced enantioselectivity to 75% ee, and lowering the temperature did not significantly increase the ee value (**4e**). Electron-withdrawing groups such as halo (F, Cl, Br) and trifluoromethoxy (CF_3O) at the *para* position of the phenyl group worked well in this process and generated the desired products in high yields with high ee values (**4f**–**4i**). It is noteworthy that CF_3O-containing molecules are significant for pharmaceuticals and agrochemicals. More electron-deficient ester and trifluoromethyl groups significantly reduced reaction activity and delivered target products in 86% ee, and 82% ee at higher temperatures, respectively (**4j, 4k**). In addition, substituents in the *meta* position also gave good results (**4l, 4m**). Surprisingly, 2-thiophene as a heterocycle was compatible with the reaction conditions to afford the corresponding product in 99% yield with 89% ee (**4n**). What excited us was that aliphatic substituted allenols were also well tolerated. Alkyl groups such as *n*-hexyl and functionalized aliphatic substituents gave generally satisfactory results (**4o**–**4r**), although product **4r** has a lower 53% yield and 81% ee. Then, allenols bearing various substituted cyclobutanol moieties were also studied. Phenyl and various aliphatic side chains proceeded smoothly and generated the products in high yields and good enantioselectivities for major diastereoisomers (**4s**–**4x**). Especially, **2x** was separated as a single *cis* isomer and delivered the product **4x** in 82% yield and 88% ee with >20:1 dr, which meant that 1,2-shift proceeded regioselectively with one of the two enantiotopic C–C bonds at cyclobutanol moiety through a desymmetrization process[62]. Diethyl substituted cyclobutanol gave a disappointing result, and only 60% ee was obtained (**4y**). Finally, different selenylating reagents were tested, and all products were obtained in excellent yields and good to excellent enantioselectivities (**4z**–**4af**). It should be noted that the absolute configurations of both **4c** and **4j** were determined to be (*S*), and the major diastereoisomers of **4u** and **4x** were determined to be (*S*, *R*) by X-ray crystallography.

Mechanistic studies

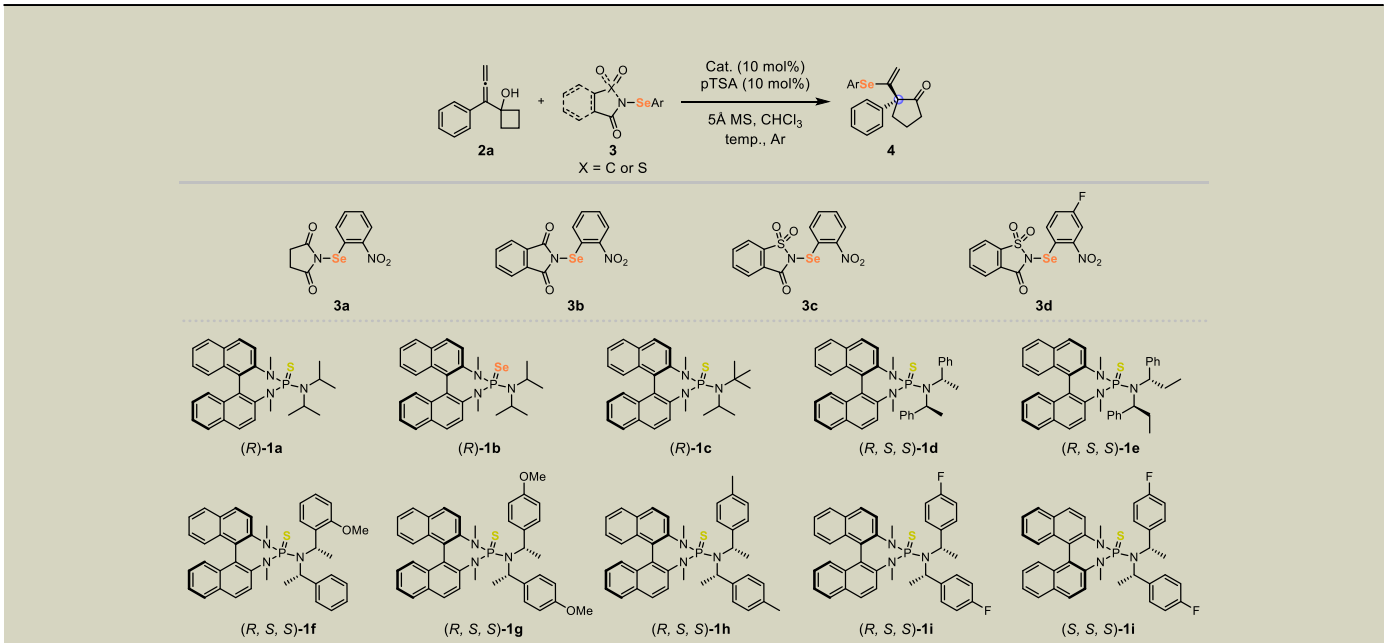
Various other types of allenol substrates have been investigated, yielding disappointing results (Fig. 3a). For instance, cyclopropanol **5a** was smoothly converted into cyclobutanone **6a**, but with low enantioselectivity. Cyclopentanol **5b** exhibited inertness under the standard conditions, possibly due to a reduction in ring strain. Monosubstituted allenol **5c** displayed reduced reactivity and was prone to decomposition under these conditions, resulting in only a 9% yield of the furan product **6c** at 0 °C. Additionally, trisubstituted allenol (*rac*)-**5d** produced the analogous furan product **6d** in 91% yield with 2% ee at −10 °C. These findings indicate that reactivity and selectivity are influenced by the number, type, and position of substituents on the allene moiety.

Next, control experiments were carried out to elucidate the mechanism of this catalytic asymmetric selenylation/semipinacol rearrangement reaction (Fig. 3b). The omission of Lewis base or acid significantly suppressed the reaction, resulting in only 18% ee of **4a** in the absence of acid. We hypothesized that the generation of catalytically active species requires the combination of Lewis base and acid in this reaction. Additionally, the counteranion of the acid was assumed to be involved in the enantiodetermining step. To verify this conjecture, a wide range of different acids were carefully tested. We found that some other sulfonic acids, such as methylsulfonic acid (MsOH), triflic acid (TfOH), (*R*)-1,1′-binaphthyl-2,2′-disulfonic acid ((*R*)-BINSA), and camphorsulfonic acid (CSA) resulted in slightly lower enantioselectivities, while diphenyl hydrogen phosphate ($(PhO)_2P(O)OH$) and trifluoroacetic acid (TFA) significantly decreased the enantioselectivities. These results indicated that sulfonic acid anion is an indispensable factor in controlling the enantioselectivity. We also observed that CSA, $(PhO)_2P(O)OH$, and TFA significantly decreased the reaction rate compared to relatively stronger acids. Based on these experimental findings, it can be concluded that both Lewis base and sulfonic acid jointly control the reactivity and enantioselectivity of this reaction.

To further clarify the formation conditions and potential structure of catalytically active species, nuclear magnetic resonance (NMR) spectroscopy and high-resolution mass spectrometry (HRMS) experiments were conducted (Fig. 3c; for details, please refer to Supplementary Information). When **3d** and (*R*, *S*, *S*)-**1i** were mixed in a 1.0:1.0 ratio in deuterochloroform ($CDCl_3$) in the absence of MsOH, no new resonances were detected in both the ^{77}Se NMR and the ^{31}P NMR spectra. As anticipated, upon adding MsOH to the above system, new signals were observed through ^{77}Se NMR (δ 700.51 ppm) and ^{31}P NMR

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Table 1 | Reaction optimization[a]



Entry	[ArSe⁺]	Cat.	Temp. (°C)	yield (%)	ee (%)
1	**3a**	(*R*)-**1a**	−10	40	81
2	**3b**	(*R*)-**1a**	−10	NR	—
3	**3c**	(*R*)-**1a**	−10	99	83
4	**3c**	(*R*)-**1a**	−40	88	84
5	**3c**	(*R*)-**1b**	−40	trace	—
6	**3c**	(*R*)-**1c**	−40	32	84
7	**3c**	(*R, S, S*)-**1d**	−40	70	83
8	**3c**	(*R, S, S*)-**1e**	−40	55	86
9	**3c**	(*R, S, S*)-**1f**	−40	35	85
10	**3c**	(*R, S, S*)-**1g**	−40	52	86
11	**3c**	(*R, S, S*)-**1h**	-40	55	85
12	**3c**	(*R, S, S*)-**1i**	−40	80	86
13	**3d**	(*R, S, S*)-**1i**	−40	94	92
14[b]	**3d**	(*R, S, S*)-**1i**	−40	98	94
15[b]	**3d**	(*S, S, S*)-**1i**	−40	98	-93

pTSA *p*-toluenesulfonic acid, *MS* molecular sieve, *2-NSA* 2-naphthalenesulfonic acid, *SFC* supercritical fluid chromatography, *HPLC* high-performance liquid chromatography, *NR* no reaction.
[a]Reaction conditions: unless otherwise noted, the reaction was conducted with **2a** (0.05 mmol), **3** (0.05 mmol), **Cat.** (0.005 mmol) and pTSA (0.005 mmol) and 5 Å MS (15 mg) in $CHCl_3$ (1.0 mL) for 60 h under Ar. Isolated yields are shown. The ee values were determined by SFC or HPLC.
[b]2-NSA (0.005 mmol) was used instead of pTSA.

(δ 70.38 ppm) spectroscopic analyses, indicating the formation of a new species. Additionally, white solid precipitation was noticed in this system after 1 h, which was confirmed to be saccharin by the ^{1}H NMR spectrum. The results of HRMS experiments suggested that the new species is most likely the ionized complex depicted in Fig. 3c [40]. This intermediate can react smoothly with allenol **2a** to generate the desired product **4a**. The relationship between the ee of the catalyst (*R, S, S*)-**1i** and the ee of the product **4a** was also explored, revealing a linear correlation (Fig. 3d). The result indicated that the catalytically active species contained one chiral catalyst and aggregates of the catalyst were not formed in this reaction[63,64]. Consequently, we consider **Int-1** to be a catalytically active intermediate in our actual reaction system (Fig. 4).

Furthermore, linear free energy relationships (LFERs) were used to identify the enantiodetermining step of the selenylation/semipinacol rearrangement reaction. We observed a good linear plot of log (*S*/*R*) (the enantioselectivities of products **4**) against σ_p (the Hammett electronic parameters of various *para*-substituted selenylating reagents) under standard conditions (Fig. 3e)[65,66]. It showed that enantioselectivity increased with the incorporation of electron-withdrawing substituents ($\rho = 1.00$). This suggests that electrophilic addition likely constitutes the enantiodetermining step.

Proposed mechanism

To better understand the origin of the enantioselectivity in this electrophilic selenylation, we performed DFT calculations at the M06-2X/6-311 + + G(d,p)+SDD(Se)-SMD($CHCl_3$)//M06-2X/6-31G(d) +LANL2DZ(Se) level of theory[67–74] (Fig. 4; for details, please refer to Supplementary Information and Supplementary Data 1). All energies discussed are Gibbs-free relative energies at 233.15 K and 1 atm in kcal/mol. The reaction begins with the formation of a complex **Cp-1** involving the catalyst (*R, S, S*)-**1i** and 2-NSA activated selenylating reagent through a chalcogen-bonding interaction between the sulfur atom of the catalyst (*R, S, S*)-**1i** and the selenium atom of **3d**[75,76]. Then,

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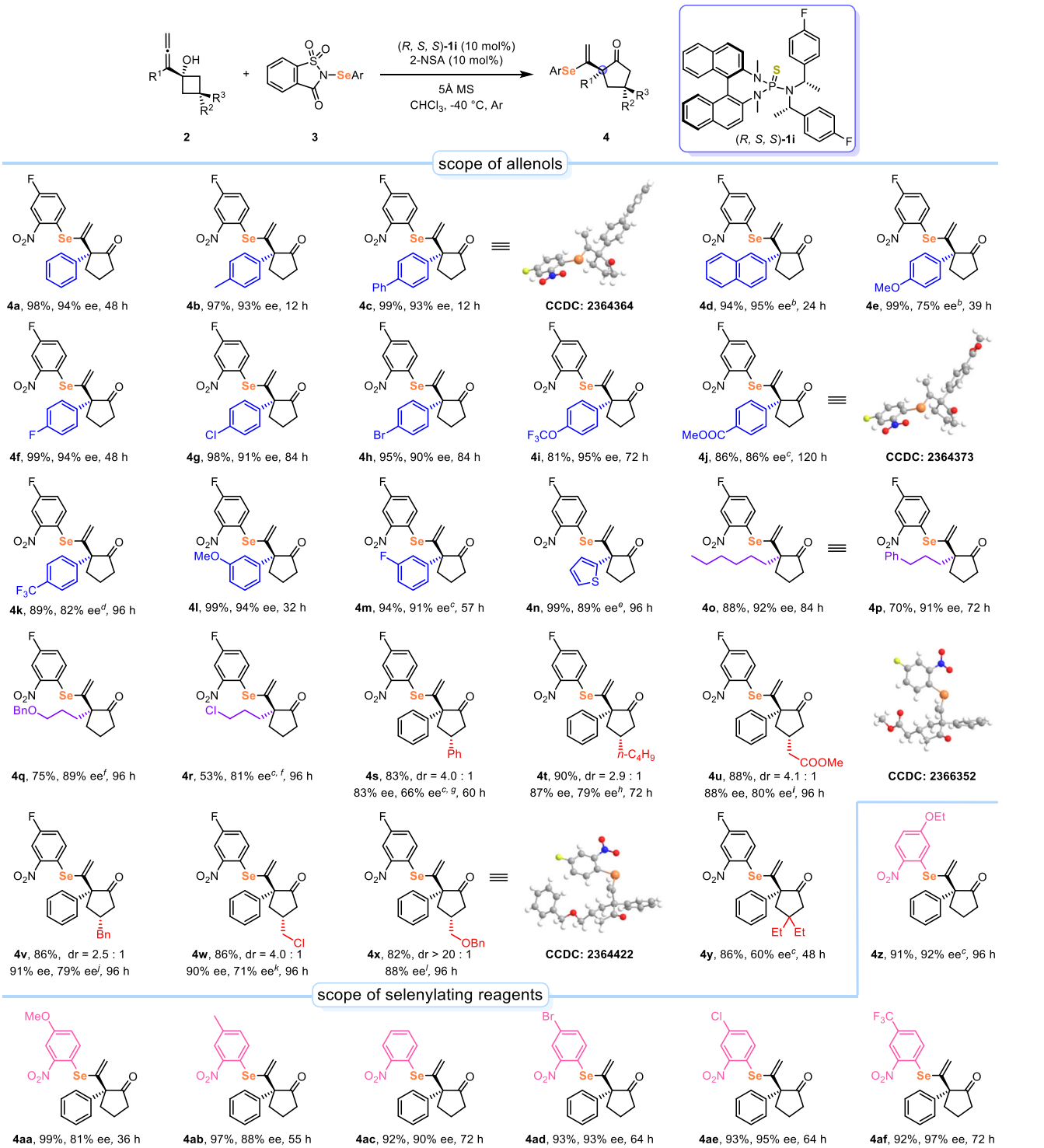
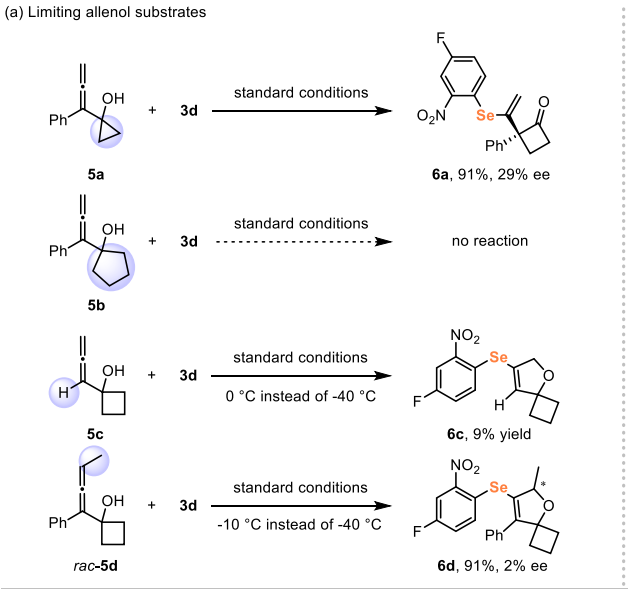


Fig. 2 | Scope of this reaction[a]. [a]Reaction conditions: unless otherwise noted, the reaction was conducted with **2** (0.1 mmol), **3** (0.1 mmol), (*R, S, S*)-**1i** (0.01 mmol), 2-NSA (0.01 mmol), and 5 Å MS (30 mg) in $CHCl_3$ (2.0 mL) at −40 °C for 12–120 h under Ar. Isolated yields are shown. The ee values were determined by SFC or HPLC. [b]The reaction was performed at −50 °C. [c]The reaction was performed at −20 °C. [d]The reaction was performed at −10 °C. [e]The reaction was performed at −60 °C. [f]0.3 equiv of 2-NSA was used. [g]**2s,** *cis*:*trans* = 5.0:1. [h]**2t,** *cis*:*trans* = 2.9:1. [i]**2u,** *cis*:*trans* = 2.9:1. [j]**2v,** *cis*:*trans* = 1.7:1. [k]**2w,** *cis*:*trans* = 4.4:1. [l]**2x,** *cis*, single diastereoisomer. 2-NSA 2-naphthalenesulfonic acid, MS molecular sieve, Bn benzyl, SFC supercritical fluid chromatography, HPLC high-performance liquid chromatography.

the activated reagent transfers the 2-nitrophenylseleneyl to the Lewis base catalyst, forming the catalytically active intermediate **Int-1**, which has been verified experimentally. Subsequently, electrophilic addition occurs to allenol **2a** via **TS-1**; the calculated activation Gibbs free energy is 13.0 kcal/mol, generating seleniranium ion **Int-2**. The final step involves a facile ring expansion by 1,2 shift and nucleophilic ring opening of the seleniranium ion **Int-2** via **TS-2**, with a barrier of only 0.4 kcal/mol relative to **Int-2** (Fig. 4a), producing the desired product **4a**. Clearly, the electrophilic addition is the enantiodetermining step, which agrees well with Hammett plot. The approach

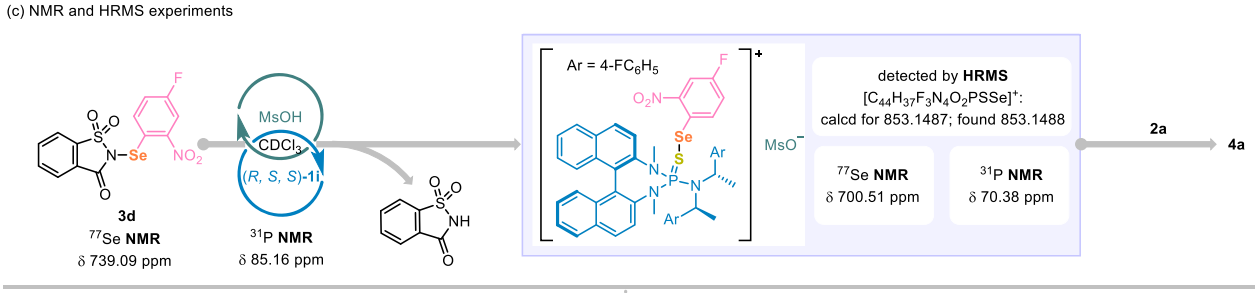
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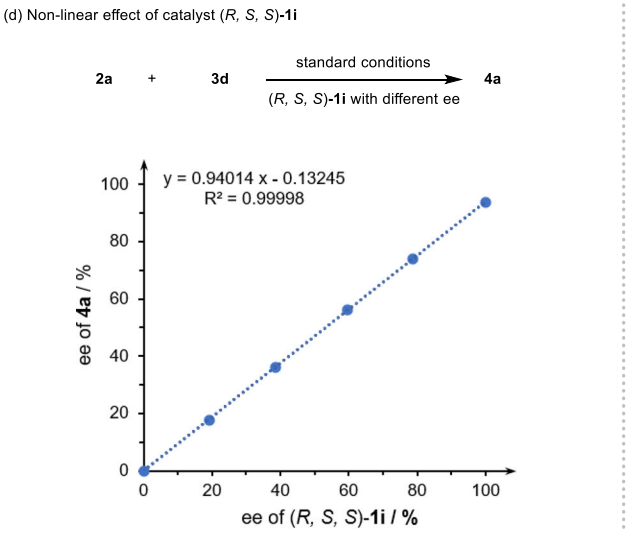


(b) The effects of catalyst and acid

2a + 3d → 4a: (*R*, *S*, *S*)-**1i** (10 mol%), acid (10 mol%), 5Å MS, $CHCl_3$, -40 °C

entry	catalyst	acid	yield (%)	ee (%)	pKa (acid)
1	(*R*, *S*, *S*)-**1i**	2-NSA	98	94	
2	—	2-NSA	10	—	
3	(*R*, *S*, *S*)-**1i**	—	5	18	
4	(*R*, *S*, *S*)-**1i**	MsOH	95	89	-1.92 (H_2O)
5	(*R*, *S*, *S*)-**1i**	(*R*)-BINSA	98	84	
6	(*R*, *S*, *S*)-**1i**	(-)-CSA	17	84	1.20 (H_2O)
7	(*R*, *S*, *S*)-**1i**	(+)-CSA	20	79	1.20 (H_2O)
8	(*R*, *S*, *S*)-**1i**	TfOH	71	89	-5.90 (H_2O)
9	(*R*, *S*, *S*)-**1i**	$(PhO)_2P(O)OH$	6	26	1.90 (H_2O)
10	(*R*, *S*, *S*)-**1i**	TFA	28	71	0.52 (H_2O)





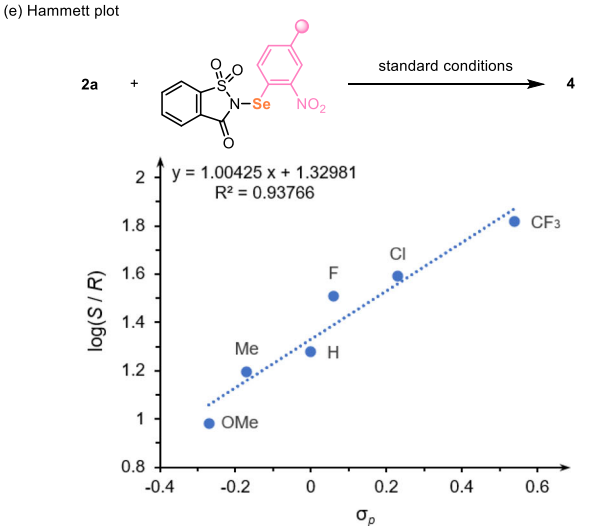


Fig. 3 | Limitations and mechanistic studies. a Limiting allenol substrates. **b** The effects of catalyst and acid. **c** NMR and HRMS experiments. **d** Non-linear effect of catalyst (*R*, *S*, *S*)-**1i**. **e** Hammett plot. 2-NSA 2-naphthalenesulfonic acid, MS molecular sieve, MsOH methylsulfonic acid, BINSA 1,1′-binaphthyl-2,2′-disulfonic acid, CSA camphorsulfonic acid, TfOH triflic acid, TFA trifluoroacetic acid, $(PhO)_2P(O)OH$ diphenyl hydrogen phosphate, NMR nuclear magnetic resonance, HRMS high-resolution mass spectrometry.

of the active intermediate **Int-1** from both the *Si* and *Re*-face of allenol **2a** was considered, and the corresponding transition states were designated as **TS-1** and **TS-1′**, respectively (Fig. 4b). The energy of computed transition structure **TS-1** leading to the major enantiomer of **4a** is 2.2 kcal/mol lower than that of **TS-1′** (13.0 vs. 15.2 kcal/mol), which is in accord with the experimentally observed enantioselectivity. Analysis of the transition state structures showed that **TS-1** features a stabilizing π···π interaction between the phenyl rings of **2a** and aromatic ring of seleniranium and four hydrogen bond interactions between the oxygen atom of sulfonic acid anion with the two hydrogen atoms of catalyst (*R*, *S*, *S*)-**1i** and two hydrogen atoms of allenol **2a** (Fig. 4b left, green dashed line). Conversely, in the disfavored transition state **TS-1′**, only the hydrogen bond interactions between the oxygen atom of sulfonic acid anion and the cyclobutanol of **2a** (Fig. 4b right, green dashed line). Consequently, the presence of more stabilizing C–H···O hydrogen bond interactions and π···π interaction results in significant stabilization of **TS-1** over **TS-1′** and are responsible for the observed enantioselectivity.

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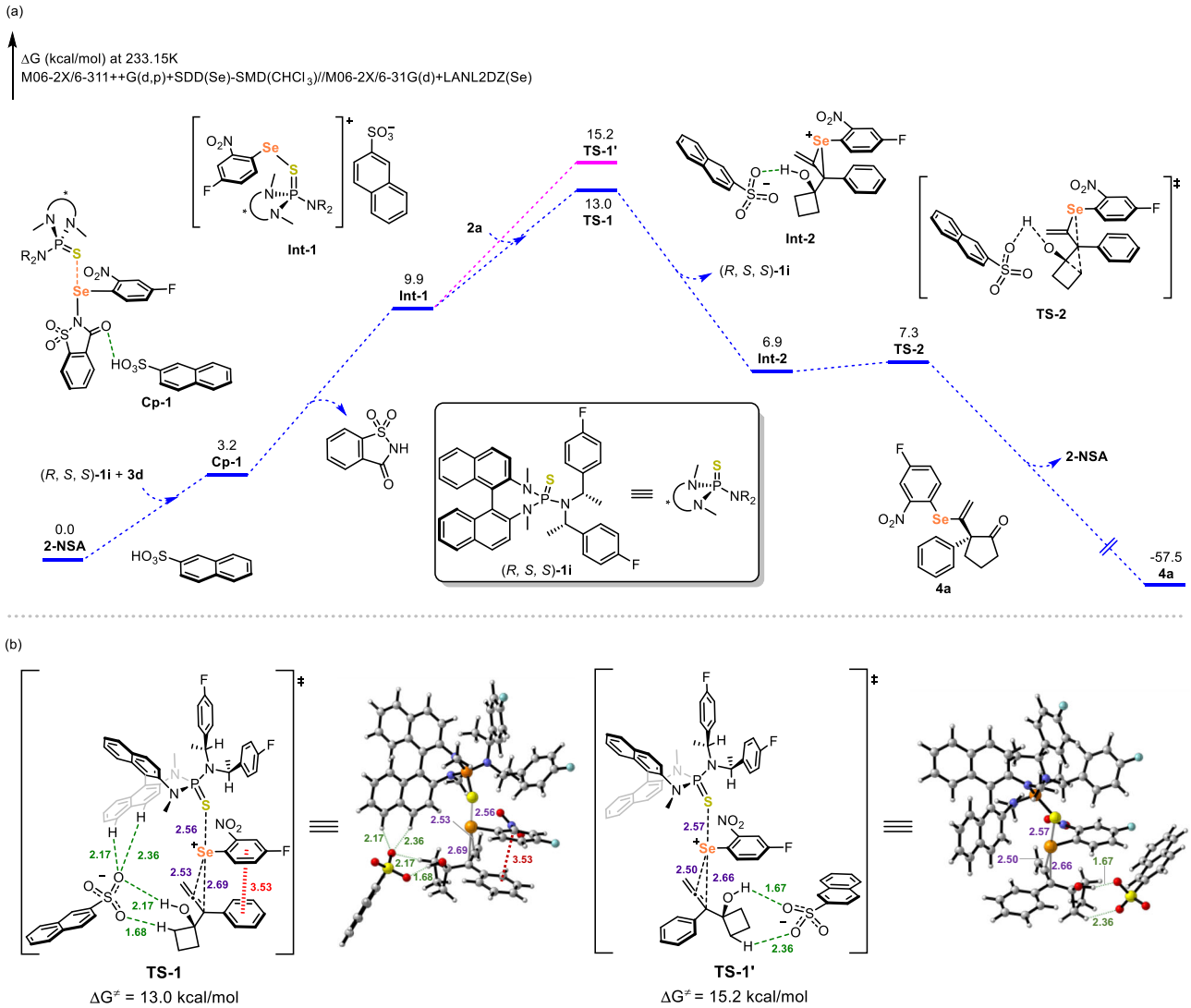


Fig. 4 | Computational studies on the reaction pathway and the origin of enantioselectivity. a Gibbs free energies profiles for the enantioselective electrophilic selenylation/semipinacol rearrangement of allenols. Orange dashed line: chalcogen-bonding interaction. Green dashed line: hydrogen-bonding interaction. Red dashed line: π···π interaction. **b** Structures analysis of transition states. Bond lengths are given in Å. Gibbs free energies are shown in kcal/mol. 2-NSA 2-naphthalenesulfonic acid.

Gram-scale reaction and synthetic applications

The gram-scale reaction and diversified product derivatization were conducted to demonstrate the practicality of this protocol (Fig. 5). Firstly, substrate **2a** (1.12 g, 6.0 mmol) smoothly reacted with **3d** under standard conditions, yielding the desired product **4a** in 98% yield with 94% ee (Fig. 5a). Subsequently, modifications were made to the vinyl selenide, nitro, and carbonyl moieties of **4a** (Fig. 5b). The vinyl selenide moiety was readily oxidized to a vinyl selenoxide which underwent rapid elimination to afford terminal alkyne **7** in 83% yield with 94% ee under basic conditions. The resulting enantioenriched alkyne bearing an α-quaternary stereocenter is considered a valuable synthon that had not been easily obtained previously. Two representative transformations of compound **7** included Sonogashira coupling and Baeyer−Villiger oxidation, both producing corresponding products without any loss in enantiopurity. Additionally, the substitution of the arylselenyl group by the bromo group using 1,3-dibromo-5,5-dimethylhydantoin (DBDMH) resulted in synthetically useful vinyl bromide **10** being afforded in moderate yield; this can be further transformed into other compounds if needed. When a significant excess of *N*-bromosuccinimide (NBS) was used, a tribromo compound **11** was formed in moderate yield, and the structure was unambiguously confirmed by X-ray crystallography. The enantioselectivities of both compounds were maintained. We then turned our attention to the aromatic nitro group, which could be selectively reduced with Zn-AcOH to provide the aniline derivative **12** in high yield without any loss in enantioselectivity. Subsequently, further transformations of **12** were investigated. The intramolecular imidization of **12** in the presence of the activating agent trimethylsilyl trifluoromethanesulfonate (TMSOTf) resulted in the desired imine **13**[77]. It could also be treated with bis(-trifluoromethyl)phenyl isothiocyanate, and furnishing chiral thiourea derivative **14** in 70% yield with 94% ee. For the carbonyl unit, Sodium borohydride ($NaBH_4$) could selectively reduce the carbonyl group of **4a** without affecting other functional groups to deliver cyclopentanol **15** in a 2.4:1 ratio of separable diastereomers (the absolute configuration of major diastereomer was determined to be (*S*, *R*) by X-ray crystallography) and maintained the enantioselectivities. The relatively low diastereoselectivity may be caused by similar steric hindrance on the *Re*/*Si*-face of the carbonyl group. Furthermore, **4a** was converted to the corresponding oxime **16** as a single *E*-isomer in 97% yield and 94% ee due to the steric hindrance.

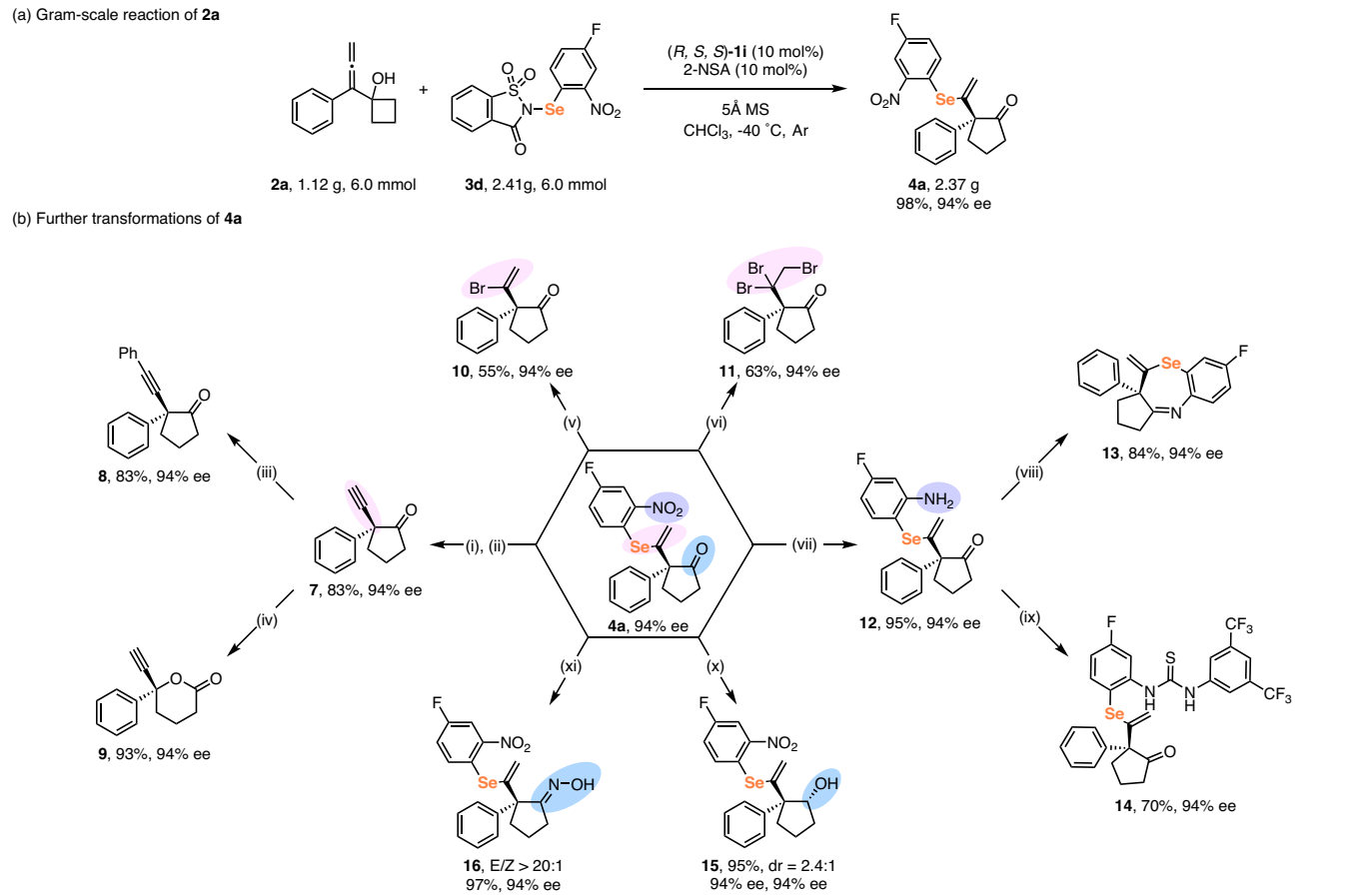


Fig. 5 | Gram-scale reaction and synthetic applications. a Gram-scale reaction of **2a**. Reaction conditions: **2a** (6.0 mmol), **3d** (6.0 mmol), (*R, S, S*)-**1i** (0.6 mmol), 2-NSA (0.6 mmol), and 5 Å MS (1.20 g) in $CHCl_3$ (80 mL) at −40 °C for 70 h under Ar. **b** Further transformations of **4a**. Reaction conditions: (i) 85% *m*-CPBA (1.1 equiv), CH_2Cl_2, 0 °C, 3 min; (ii) DABCO (1.0 equiv), toluene, rt to 90 °C, 8 min; (iii) PhI (1.2 equiv), $Pd(PPh_3)_2Cl_2$ (0.02 equiv), CuI (0.04 equiv), Et_3N, rt, 11 h; (iv) MMPP·$6H_2O$ (2.0 equiv), $NaHCO_3$ (3.0 equiv), MeOH, rt, 12 h; (v) DBDMH (1.0 equiv), CH_2Cl_2, rt, 11 h; (vi) NBS (6.0 equiv), CH_2Cl_2, rt, 10 h; (vii) Zn (5.0 equiv), AcOH/THF/H_2O, 60 °C, 1 h; (viii) TMSOTf (2.5 equiv), DMF, 70 °C, 3 h; (ix) 3,5-$CF_3C_6H_3N=C=S$ (1.1 equiv), THF, rt, 24 h; (x) $NaBH_4$ (1.5 equiv), MeOH, 0 °C, 1 h; (xi) NH_2OH·HCl (2.0 equiv), pyridine, 90 °C, 36 h. Isolated yields are shown. The ee values were determined by SFC. *m*-CPBA *m*-chloroperbenzoic acid, DABCO 1,4-diazabicyclo[2.2.2]octane, MMPP magnesium monoperoxyphthalate, DBDMH 1,3-dibromo-5,5-dimethylhydantoin, NBS *N*-bromosuccinimide, AcOH acetic acid, THF tetrahydrofuran, TMSOTf trimethylsilyl trifluoromethanesulfonate, DMF *N,N*-dimethylformamide. SFC supercritical fluid chromatography.

Discussion

In conclusion, we have developed a highly enantioselective electrophilic selenylation/semipinacol rearrangement of allenols facilitated by the cooperative catalysis of a chiral sulfide and an achiral sulfonic acid. Various cyclopentanones containing an arylselenovinyl-substituted quaternary carbon stereocenter were obtained in high yields and enantioselectivities, which were further transformed into synthetically valuable alkyne, vinyl bromide, and aniline derivatives. Mechanistic experiments and DFT calculations indicated that both the chiral Lewis base and the achiral sulfonic acid jointly regulate the reactivity and enantioselectivity of the reaction, with the electrophilic addition step determined to be the enantiodetermining step. Furthermore, these effective catalysts and selenylating reagents are being employed for other selenofunctionalization reactions in our laboratory.

Methods

General procedure for the enantioselective electrophilic selenylation/semipinacol rearrangement of allenols

An oven-dried tube charged with **3** (0.10 mmol, 1.0 equiv), (*R*, *S*, *S*)-**1i** (6.3 mg, 0.01 mmol, 0.1 equiv), 2-NSA (2.1 mg, 0.01 mmol, 0.1 equiv) and activated 5 Å MS (30.0 mg) was evacuated and backfilled with argon three times. Subsequently, a solution of **2** (0.10 mmol, 1.0 equiv) in distilled $CHCl_3$ (2.0 mL) was added to the reaction system at the corresponding temperature. The resulting mixture continued to stir for 0.5–5 days. After the reaction was complete, direct purification by column chromatography on silica gel afforded the corresponding product **4**. The ee was determined by HPLC or SFC.

Data availability

The data generated in this study are provided in the Supplementary Information file. The experimental procedures and data of NMR, HRMS, SFC, and HPLC have been deposited in the Supplementary Information file. The X-ray crystallographic coordinates for structures reported in this study have been deposited at the Cambridge Crystallographic Data Centre, under deposition numbers 2364364 (**4c**), 2364373 (**4j**), 2366352 (**4u**), 2364422 (**4x**), CCDC 2366354 (**11**), and CCDC 2364406 (**15**). These data could be obtained free of charge from The Cambridge Crystallographic Data Centre (https://www.ccdc.cam.ac.uk/data_request/cif). All data are available from the corresponding author upon request.

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Author contributions

R.-F.C. and Z.-M.C. proposed and designed experiments. R.-F.C. completed the reaction optimization, substrate scope exploration, mechanistic studies, and synthetic applications. Z.-W.W. and Z.-L.L. helped synthesize various substrates. D.Z. and Y.-X.H. helped analyze the data. R.S. and X.-S.X. achieved DFT calculations. R.-F.C., R.S., X.-S.X., and Z.-M.C. wrote the manuscript. Z.-M.C. directed the project.

Competing interests

The authors declare no competing interests.

Additional information

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